WASIM ANSARI

AI / Machine Learning Engineer | Generative AI & Agentic Systems | LLM Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on, production experience designing and deploying agentic AI systems, LLM-powered applications, and end-to-end machine learning pipelines using Python, LangChain/LangGraph, RAG architectures, and MLOps tooling (FastAPI, Docker, CI/CD, cloud platforms). Brings a distinctive combination of deep technical execution and 8+ years of cross-industry business experience across legal services, real estate, and international trade – translating into strong product intuition and the ability to design AI systems that solve real, measurable business problems.

CORE TECHNICAL SKILLS

- **Generative AI & Agentic Systems:** LangChain, LangGraph, CrewAI, Agno, MCP (Model Context Protocol) clients & servers, Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Orchestration
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Neural Networks, Computer Vision, Transformer Fine-Tuning, Scikit-learn, PyTorch, PyTorch Lightning, Keras, HuggingFace Transformers, XGBoost
- **LLM Infrastructure & Retrieval:** ChromaDB, FAISS, Neo4j (hybrid graph + vector retrieval), Redis, Celery, Pydantic
- **MLOps & Deployment:** FastAPI, Docker, Git/GitHub Actions (CI/CD), Google Cloud Run, Cloud SQL, Cloud Build, Azure, Vercel, Hugging Face Spaces
- **Data Analysis & Visualization:** Python, Pandas, NumPy, Statistical & Hypothesis Testing, Matplotlib, Seaborn, Power BI
- **Monitoring, Automation & Observability:** WandB, TrackIO, LangSmith, Guardrailing, Prompt Experimentation, n8n Workflow Automation

FLAGSHIP PROJECT

HyreFast – AI-Powered Recruiting Platform *(Intern & Associate AI Engineer at TechPranee)*

- **Production Architecture:** Designed and built a full-stack AI recruiting platform on LangGraph, FastAPI, Neo4j, Redis, Celery, Pydantic, Ollama, and MCP – covering resume ingestion, structured extraction, and candidate-role matching end-to-end.
- **Resume Extraction Pipeline:** Built a document parsing pipeline (PyMuPDF/pdfplumber + LLM-based structuring) with rule-based fallback and Pydantic schema validation to guarantee deterministic, production-reliable output.
- **Self-Healing Ingestion:** Engineered a candidate ingestion pipeline with a classified-error repair loop and history-aware retry prompting, significantly improving structured-extraction reliability under real-world resume variance.
- **Skill Normalization at Scale:** Built a multi-stage skill-matching pipeline (exact lookup → BM25 full-text search → lexical scoring → synonym expansion → graph traversal) against a self-authored subcategory skill taxonomy.
- **Graph-Based Retrieval:** Architected a Neo4j graph optimized for 1-2 hop traversal, powering hybrid keyword, vector, and graph retrieval for candidate-job matching.
- **Technical Leadership:** Authored architecture documentation and onboarded a junior engineer onto the codebase, establishing reusable engineering patterns for the platform.

PROFESSIONAL EXPERIENCE

AI/ML Engineering Intern | TechPranee (SpearHub) *Dec 2025 – Jun 2026*

- Designed, developed, and deployed production-ready AI applications using machine learning and generative AI to automate business workflows and improve operational efficiency.
- Built and deployed a time-series forecasting model integrated into ERP systems, and a hybrid RAG chatbot (Neo4j graph + vector search) for HR screening and decision automation.
- Delivered multiple multi-agent AI applications integrated with existing business systems, leading rapid experimentation cycles and presenting actionable recommendations to project leadership.
- Optimized system performance for high-load, concurrent production usage; owned debugging, testing, and continuous improvement of deployed systems to maintain stability at scale.

AI/ML Engineer & GenAI Architect | WSMAISYS (Independent Freelancer) *Jan 2025 – Dec 2025*

- Delivered freelance AI/ML solutions for independent clients, including predictive modeling, AI-powered tools, and workflow automation using n8n.
- Built a diverse portfolio of end-to-end AI/ML projects (see below) demonstrating strong problem-solving ability and technical versatility across the ML/GenAI stack.
- Pursued an IIT Madras Diploma in Data Science with focus on Deep Learning, Generative AI, Reinforcement Learning and cloud-based machine learning systems.

Prior Professional Experiences

8+ years of cross-industry experience preceding the AI/ML career transition – foundational domain expertise now applied directly to AI project design.

- **Retail Sales Manager, Booze Blitz – Ahaan Limited (UK), 2022–2023:** Led compliance and strategic sales planning that grew revenue by 20%; this exposure to pricing and demand patterns later shaped ML-based pricing and demand-forecasting project work.
- **Import/Export Entrepreneur, Ahlan Exports (India), 2021–2022:** Managed international B2B logistics, compliance, and digital marketing funnels – experience that informs current interest in AI-driven supply-chain optimization.
- **Real Estate Consultant, FAHZAM Properties (Dubai, UAE), 2015–2020:** Certified RERA Broker managing property portfolios and legal clearances; domain insight now applied to AI-driven real estate valuation and document-automation projects.
- **Legal Associate, QuisLex Legal Services & Research Associate, Mindcrest India, 2012–2014:** Reviewed legal documentation and classified case law with 90%+ accuracy in half the standard processing time – domain expertise that directly informed the Jurisol legal-AI assistant project below.

ADDITIONAL PROJECTS

- **MediFlow – Agentic Medical AI Assistant:** Clinical decision-support chatbot combining RAG, patient history analysis, and MCP-based tool use for nephrology workflows. Stack: LangChain, LangGraph, LangSmith, MCP, Google Cloud Run, CI/CD.
- **Jurisol – Legal AI Assistant:** RAG-based legal research assistant with ChromaDB vector search, document summarization, and live legal-document lookup. Stack: RAG, LangGraph, FastAPI, Mistral LLM.
- **BikeValuePro – Price Prediction Model:** XGBoost/Scikit-learn regression model for used-bike pricing achieving 94.9% accuracy, with a Streamlit UI for personalized pricing advice.
- **AI Travel Planner – Multi-Agent Orchestration:** Three-agent CrewAI system that autonomously researches destinations, analyzes preferences, and generates personalized itineraries.
- **IITM Assignment Solver – Autonomous Code Agent:** Sandboxed AI agent that processes ZIP/CSV/PDF/Excel/JSON inputs and generates and executes Python code; secured an A+ grade in an IIT Madras college challenge.
- **PDF Excel Extractor & DocuDroid – Document AI Tools:** LLM-driven schema detection converting unstructured PDFs into structured Excel output (BLEU-scored), and an in-memory RAG bot for instant Q&A over PDFs/webpages.

EDUCATION

Indian Institute of Technology, Madras – Diploma in Data Science (CGPA 8+) *2024 – 2026*
Coursework: Machine Learning, Linear Algebra, Statistics, Business Analytics, Deep Learning, Generative AI.

Indian Institute of Technology, Madras – Diploma in Programming *2026 - Ongoing*
Coursework: Computational Thinking, Python, Java, DBMS, DSA, Linux/Systems, Modern App Development.

Savitribai Phule Pune University – Bachelor of Laws (5-Year Integrated) *2007 – 2012*

CERTIFICATIONS & LANGUAGES

- Certifications: Understanding Google Cloud Platform | Microsoft Azure Essentials | Dubai RERA Certification
- Languages: English (IELTS 7 Band) | Hindi (Native/Bilingual) | Urdu (Native/Bilingual)

SELECTED PUBLICATIONS

- **Adapting Export Strategies for Indian Handmade Carpets:** Data-driven export strategy for India's handmade carpet industry navigating rising U.S. tariffs, using Python-led trade analytics and market opportunity scoring.
- **Teaching Computers to See Like Humans:** Explainer on how CNNs use kernels, feature maps, and layering to power modern computer-vision applications.